

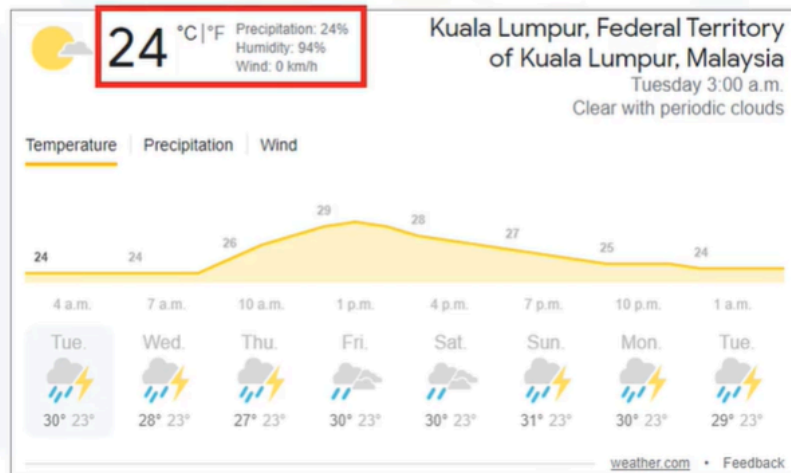
# Facts and Dimensional Modeling

## Facts and dimensions

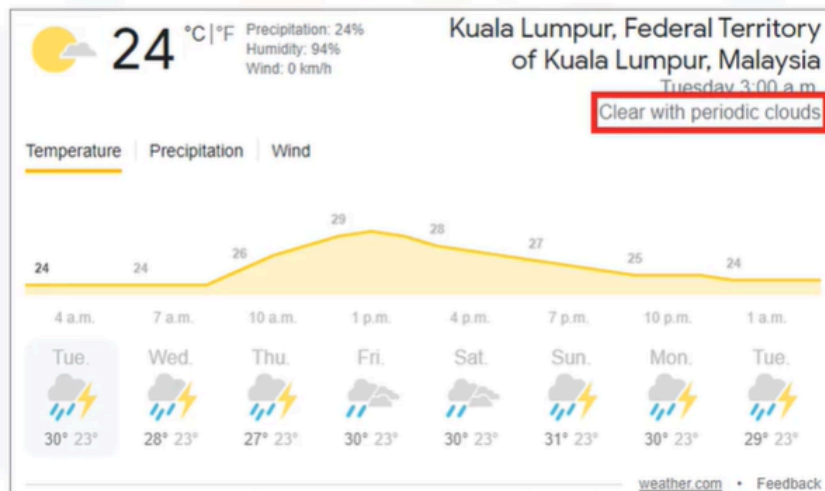
---

- Data can be categorized as facts and dimensions
  - Facts are usually measured quantities, such as temperature, number of sales, or mm of rainfall
  - Facts can also be qualitative
  - Dimensions are attributes relating to facts
  - Dimensions provide context to facts
    - "24°C" is not useful information by itself
- 
- Data can be lumped into two categories: Facts and dimensions. Facts are usually quantities which can be measured, such as temperature, number of sales, or millimeters of rainfall. But facts can also be qualitative in nature. Dimensions are attributes which can be assigned to facts. Dimensions provide context to facts, which makes facts useful. For example, a temperature such as "24 degrees celsius," all by itself, is not meaningful information.

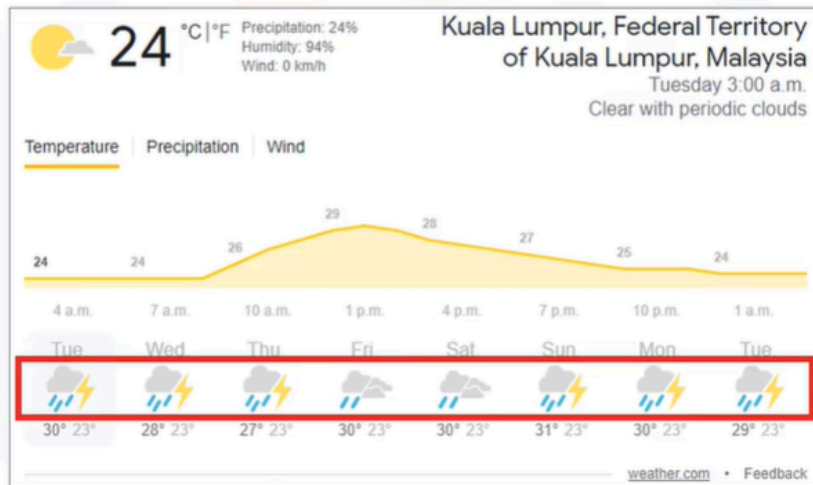
## Facts and dimensions



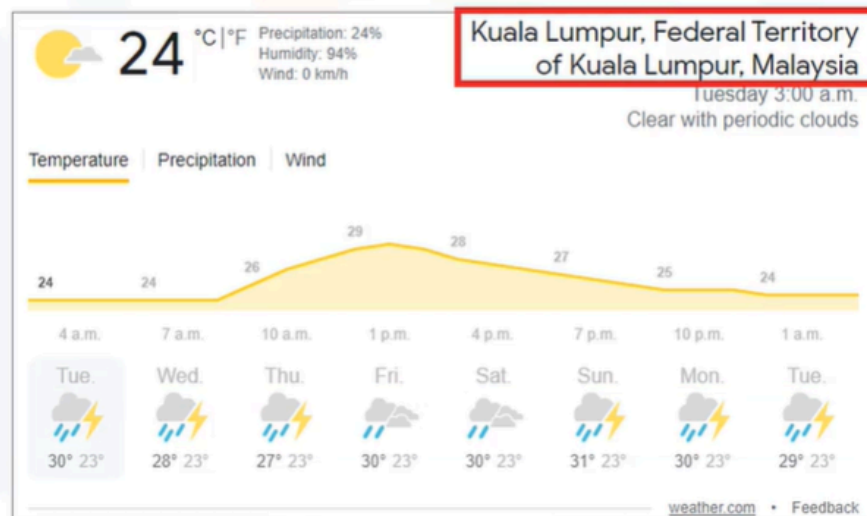
## Facts and dimensions



## Facts and dimensions



## Facts and dimensions



- Let's look at a familiar example. Here we have a weather report obtained by googling "weather in Kuala Lumpur." The facts here are things like the temperature, humidity, probability of precipitation, and wind speed. Notice that these are all quantities. But there are other facts here, such as this icon, which means "partly cloudy," or the statement "Clear with periodic clouds," and these other icons which indicate forecasted conditions, such as

thundershowers. All of these non-numeric facts are examples of qualitative facts. To make sense of these facts, we need to provide context. Context is provided by dimensions, which includes things like the location, "Kuala Lumpur, Malaysia" and the day and time, "Tuesday at 3 a.m." The statement "24°C in Kuala Lumpur, Malaysia on Tuesday, August 17th at 3:00 a.m." means something.

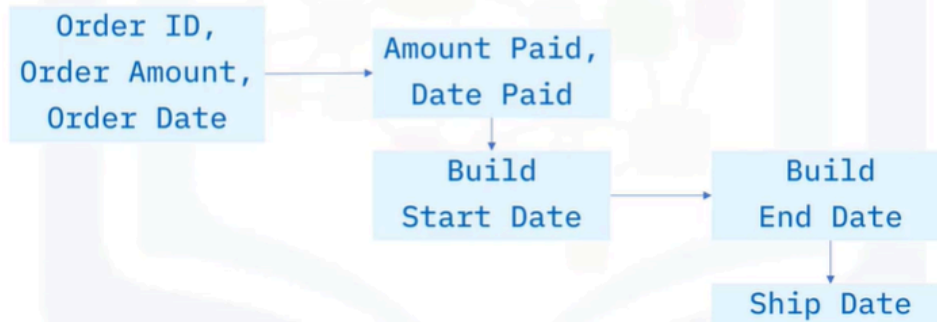
## Fact tables

---

- Facts of a business process, plus
  - Foreign keys to dimension tables
    - Dollar amounts for sales transactions
  - Can contain detail level facts, or
  - Facts that have been aggregated
  - Summary tables contain aggregated facts
    - "Quarterly Sales" summary table, with
    - "store\_id" as foreign key
- 
- A fact table typically consists of the facts of a business process, and it also contains foreign keys which establish well-defined links to dimension tables. Usually, facts are additive measures or metrics, such as dollar amounts for individual sales transactions. A fact table can contain detail level facts, such as individual sales transactions, or facts that have been aggregated, such as daily or weekly sales totals. Fact tables that contain aggregated facts are called summary tables. For example, you could summarize sales transactions by summing overall sales for each quarter of the year. An example of a foreign key might be 'store ID.'

## Accumulating snapshot fact tables

- Used to record events during a well-defined business process



- "Accumulating snapshot" fact tables are used to record events that take place during a well-defined business process. For example, suppose you have finished configuring a custom computer for yourself online, and you have just placed your order. The order date and the order amount are recorded in a snapshot table by the manufacturer. A unique "order ID" is also assigned. Once your order is verified, your payment is processed. The "amount paid" and the "date paid" are then recorded. After payment verification, the computer specifications are sent to the manufacturing department. Once the computer is in production, the "build start-date" is entered, and once the computer is built, the "build end-date" is recorded. Finally, once your computer is ready to be sent, the "ship date" is recorded. All these fields are stored in a single row of the table, which is uniquely identified by the "order ID."

## Dimensions

---

- Dimensions categorize facts
  - Called categorical variables in stats and machine learning
  - Used to answer business questions
  - Used for filtering, grouping, and labelling
    - People, product, and place names, and date or time stamps
  - Dimension table stores dimensions of a fact
    - Joined to fact table via foreign key
- 
- A dimension is a variable that categorizes facts, Dimensions are called "categorical variables" by statisticians and machine learning engineers. Dimensions enable users to answer business questions. The main uses for dimensions in analytics include filtering, grouping, and labeling operations. For example, commonly used dimensions include names of people, products, and places, and date or time stamps. A dimension table thus stores the dimensions of a fact and is joined to the fact table via a foreign key.

## Dimension table examples

---

### Product tables:

- Make, model, color, size

### Employee tables:

- Name, title, department

### Temporal tables:

- Date/time at granularity of recorded events

### Geography tables:

- Country, state, city, region





- Some examples of various types of dimension tables include: Product tables, which describe products, such as make, model, color, and size. Employee tables, which describe employees, such as name, title, and department. Temporal tables, which describe time at the level of granularity, or precision, at which events are recorded. Geography tables, for location data such as country, state, city, and postal or zip code.

## Example schema with fact & dimension tables



- Each fact table typically has multiple dimension tables related to it

- Let us look at an example of a schema with both fact and dimension tables to help illustrate their relationships. We could have a fact table for recording sales at a car dealership. This table would store facts about each car sale such as the Sale date, and Sale amount as well as a primary key "Sale ID." For each sales transaction, we also need to record its dimensions like the Vehicle sold and the Salesperson who sold it. The attributes of these dimensions, such as the vehicle Make and Model or the Salesperson's First and Last name are stored in separate Vehicle and Salesperson tables. However, we link them with the Sales table by recording the "Vehicle ID" and "Salesperson ID" as foreign keys in the Fact table. This way we typically end up with multiple dimension tables for each fact table.

Sure! Let's talk about facts and dimensions in data warehousing.

In simple terms, facts are the measurable pieces of information we collect, like the number of sales or the temperature outside. Think of facts as the numbers that tell us what happened. For example, if you sold 50 ice creams on a hot day, "50" is the fact. On the other hand, dimensions are the details that give context to those facts. They help us understand the "who," "what," "where," and "when." In our ice cream example, dimensions could include the flavor of the ice cream, the location of the sale, and the date. So, when we say, "50 vanilla ice creams sold at the beach on July 1st," the "50" is the fact, and the rest are dimensions that help us understand the situation better.

## What relationships exist between fact and dimension tables?

The relationship between fact and dimension tables is crucial in data warehousing. Here are the key points:

- **Fact Tables:** These tables store quantitative data (facts) about business processes, such as sales amounts or transaction counts. Each record in a fact table typically includes:
  - **Measures:** The numerical values (e.g., sales amount).
  - **Foreign Keys:** Links to dimension tables that provide context for the facts.
- **Dimension Tables:** These tables store descriptive attributes (dimensions) related to the facts. They provide context and help answer business questions. For example, a dimension table might include:
  - **Attributes:** Details like product names, customer information, or time periods.
- **Relationships:**
  - **Foreign Key Relationships:** Fact tables contain foreign keys that reference primary keys in dimension tables. This establishes a link between the two.



- **One-to-Many Relationship:** Each record in a dimension table can relate to multiple records in a fact table. For example, one product (dimension) can be sold in many transactions (facts).
- **Star Schema:** A common design where a central fact table is surrounded by multiple dimension tables, making it easy to query and analyze data.

This structure allows for efficient data analysis and reporting, as it connects quantitative data with descriptive context.